Import Packages

In [1]:
```
import pandas as pd
import numpy as np
import seaborn as sb
import scipy.stats as sp
import matplotlib.pyplot as plt
```

Load Feature Dataset (1 file)

In [3]:
```
FeatureData = pd.read_csv(
    "./data/FeatureData.csv",
    header=None,
)
FeatureData
```

Out[3]:

	0	1	2	3	4	5	6	7	8	9	...
0	1.351000	31.661000	31.832000	1.418300	1.053400	30.628000	0.992040	0.992420	1.059700	1.170800	...
1	-1.372000	-22.786000	-23.613000	-1.085600	-1.057500	-19.468000	-1.319600	-1.056400	-2.041700	-1.343200	...
2	0.011083	0.023339	0.020506	0.027215	0.016574	0.018563	0.020904	0.024480	0.029605	0.028426	...
3	0.426105	2.312749	2.313820	0.396240	0.388252	2.088591	0.403801	0.404898	0.381526	0.412919	...
4	0.181443	5.348262	5.353342	0.156266	0.150465	4.361866	0.162618	0.163343	0.144686	0.169694	...
...	...	...	...	...	...	...	...	...	...	...	...
265	-0.069441	-0.095775	-0.094919	-0.098230	-0.091625	-0.100346	-0.101488	-0.105314	-0.111518	-0.094917	...
266	-1.500603	-1.513689	-1.520788	-1.502072	-1.514240	-1.522188	-1.515150	-1.520913	-1.511260	-1.516616	...
267	1.372567	1.390663	1.361458	1.367672	1.364317	1.356064	1.355348	1.343550	1.382847	1.354853	...
268	1.136464	1.123389	1.128936	1.125794	1.126631	1.126265	1.125331	1.126091	1.121208	1.126622	...
269	1.559874	1.562256	1.536999	1.539718	1.537082	1.527288	1.525215	1.512959	1.550459	1.526407	...

270 rows × 360 columns

Declare normal & abnormal features seperately

In [4]:
```
NoOfData = int(
    FeatureData.shape[1] / 2
)  # int function: convert float to integer
Normal_FeatureData = FeatureData.iloc[:, :NoOfData]
Abnormal_FeatureData = FeatureData.iloc[:, NoOfData:]

print(Normal_FeatureData.shape)
print(Abnormal_FeatureData.shape)
```

```
(270, 180)
(270, 180)
```

P-value calculation (t-Test)

P-value (0 ~ 1) is the representative indicator of statistical difference between two or more groups.

- low p-value (< 0.05) represents that those groups are well distinguished.
- Watch the following YouTube video to better understand about p-value.

In [ ]:

```python
from IPython.display import YouTubeVideo

YouTubeVideo("YSwmpAmLV2s")
```

Out[ ]:

In [5]:

```python
NoOfFeature = FeatureData.shape[0]  # Number of feature: 270

P_value = np.zeros((NoOfFeature, 2))

# t-Test (Result: p-value)
for i in np.arange(NoOfFeature):

    T_test = np.array(
        sp.ttest_ind(
            Normal_FeatureData.iloc[i, :], Abnormal_FeatureData.iloc[i, :]
        )
    )
    P_value[i, 0] = i  # Index of feature (0~269)
    P_value[i, 1] = T_test[1]  # P-value

P_value = pd.DataFrame(P_value, columns=["No.", "P-value"])
P_value
```

Out[5]:

	No.	P-value
0	0.0	1.014879e-11
1	1.0	8.509037e-11
2	2.0	2.246947e-08
3	3.0	1.340717e-15
4	4.0	5.902344e-12
...	...	...
265	265.0	1.487181e-01
266	266.0	3.428681e-01
267	267.0	4.592642e-08
268	268.0	6.636401e-21
269	269.0	1.567811e-03

270 rows × 2 columns

In [6]:

```python
# Sort by P-value in ascending order
P_value_Rank = P_value.sort_values(["P-value"], ascending=True)
P_value_Rank
```

Out[6]:

	No.	P-value
198	198.0	4.762393e-81
110	110.0	5.270311e-77
170	170.0	4.615377e-76
133	133.0	5.747675e-74
134	134.0	7.585991e-74
...	...	...
85	85.0	8.999631e-01
95	95.0	9.358453e-01
255	255.0	9.450087e-01
217	217.0	9.695416e-01
166	166.0	9.776721e-01

270 rows × 2 columns

In [7]:

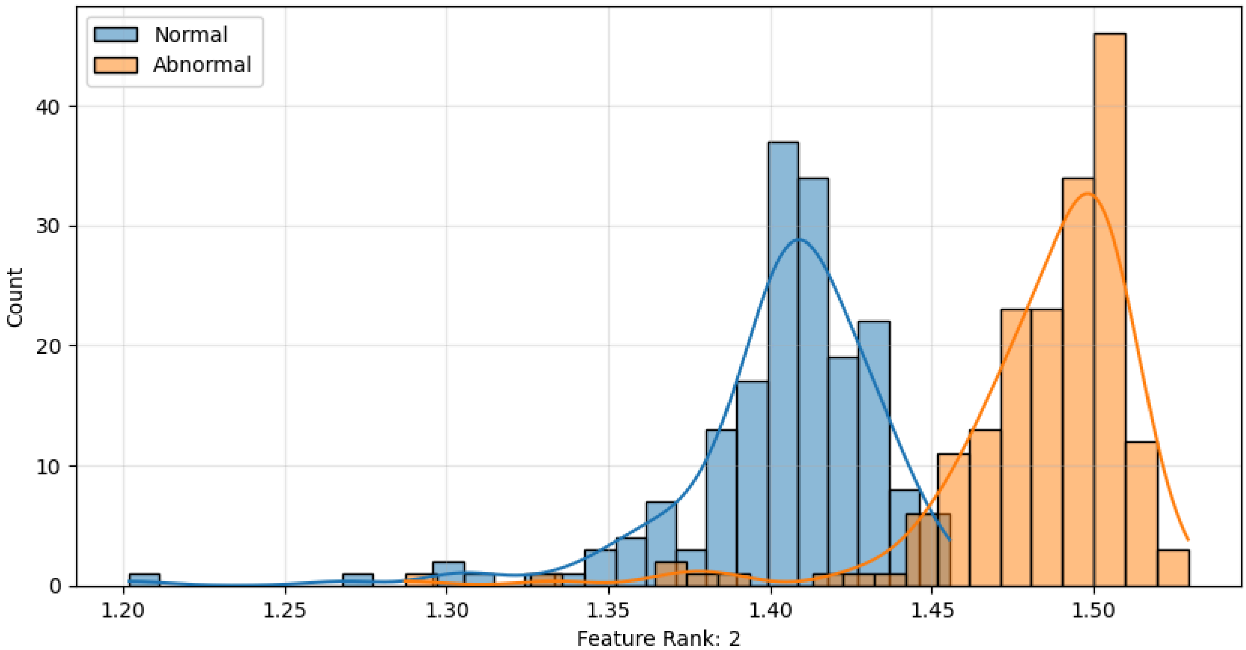
```
# Save t-Test result (p-value)
path1 = "./data/P_value.csv"
path2 = "./data/P_value_Rank.csv"
P_value.to_csv(path1, sep=",", header=None, index=None)
P_value_Rank.to_csv(path2, sep=",", header=None, index=None)
```

Confirm PDF(Probabilistic Density Function) Graphs (Normal vs Abnormal)

In [8]:

```
# P-value Rank (0 ~ 269)
FeatureRank = 1

# PDF Chart
plt.figure(figsize=(10, 5))
sb.histplot(
    Normal_FeatureData.iloc[int(P_value_Rank.iloc[FeatureRank, 0]), :],
    kde=True,
    label="Normal",
)
sb.histplot(
    Abnormal_FeatureData.iloc[int(P_value_Rank.iloc[FeatureRank, 0]), :],
    kde=True,
    label="Abnormal",
)
plt.legend(loc="best", fontsize=10)
plt.xlabel(f"Feature Rank: {FeatureRank+1}")
plt.grid(alpha=0.3)
plt.show()
```



Select Top Features Having Low P-values

In [9]:
```
Rank = 30  # Select number of rank

Normal = np.zeros((Rank, NoOfData))
Abnormal = np.zeros((Rank, NoOfData))

for i in range(Rank):

    index = int(P_value_Rank.iloc[i, 0])
    Normal[i, :] = Normal_FeatureData.iloc[index, :].values
    Abnormal[i, :] = Abnormal_FeatureData.iloc[index, :].values

FeatureSelected = pd.DataFrame(np.concatenate([Normal, Abnormal], axis=1))

print("Selected Feature Data Size :", FeatureSelected.shape)
```

```
Selected Feature Data Size : (30, 360)
```

Save Finally Selected Feature Data in Drive (.csv)

In [10]:
```
path = "./data/FeatureSelected.csv"
FeatureSelected.to_csv(path, sep=",", header=None, index=None)
```